

Ocean LCL Multi-Commodity Routing with Consolidation

MVP Formal Model — Phase 1, v1

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Draft — 2026-05-16

Abstract

Formal specification of the optimization model for the ocean LCL (Less-than-Container-Load) multi-commodity routing problem with bin-packing-based consolidation. The model is a *joint* Binary Multi-Commodity Flow + bin-packing problem: simultaneously decide which LCL shipments share which container, which container type (FEU/TEU) to use, and which NVOCC sailing carries the consolidated container. The model parallels the Ocean FCL formulation (`model/ocean_fcl_routing.tex`) and Air Freight formulation (`model/air_freight_routing.tex`) for notational consistency. A sequential decomposition solution strategy is documented (§8) for tractability at production scale, where the joint problem becomes prohibitive beyond ~50–100 shipments per batch. The model itself is specified in joint form to preserve correctness; decomposition is a solving technique, not a model simplification.

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1 Problem Statement

Given a set of LCL shipment requests (commodities), each defined by origin location, destination location, cargo volume, cargo weight, and time constraints, find a minimum-cost joint assignment of:

1. Each shipment to one shared container,
2. Each container to one container type (FEU or TEU),
3. Each container to one NVOCC sailing,

such that:

- every shipment is delivered on or before its deadline,
- each container's contents respect volume and weight capacity,
- each NVOCC sailing's container capacity is respected,
- co-loaded shipments are compatible (hazmat segregation),
- CFS cutoffs and deadlines are honored,
- optional per-shipment budget caps are respected.

NVOCC operating model. A freight forwarder acting as a Non-Vessel-Operating Common Carrier (NVOCC) buys FCL container capacity from ocean carriers (per-container rates or BSA) and resells space inside those containers to LCL shippers at per-CBM or weight/measurement (W/M) rates. The cost minimization in this model is on the *carrier-facing* side: minimize the number and cost of FCL containers purchased to carry all LCL shipments. Customer-facing pricing (W/M rates to shippers) is out of scope; see `PRD.md §6.3` for the pricing engine roadmap.

MVP scope.

- Single-shipment routing per LCL shipment; consolidation across shipments is the joint decision.
- Container types: FEU (40'HC, 76 CBM usable, 26,500 kg payload) and TEU (20' standard, 33 CBM, 24,000 kg). Optimizer selects mix.
- NVOCC sailings: weekly cadence on major lanes; modeled as discrete sailing arcs with CFS cutoff and ETD.
- Origin CFS and destination CFS as explicit nodes: consolidation (build-up) at origin, deconsolidation (breakdown) at destination. CFS dwell times included.
- Container fill rate cap $\eta \approx 0.97$ (no perfectly packed containers in practice).
- No transshipment in MVP (single-leg NVOCC sailing only).
- Hazmat compatibility: pairwise exclusion matrix; full DGR segregation rules deferred.
- Probabilistic transit time objectives deferred to P1.
- Sequential-decomposition solving documented; joint model preserved for small batches and correctness validation.

2 Graph Structure

2.1 Physical Network Graph $G(N, A)$

The LCL freight network is a directed graph $G = (N, A)$.

$$N = N_O \cup N_{\text{CFS},O} \cup N_{\text{POL}} \cup N_{\text{POD},A} \cup N_{\text{POD},E} \cup N_{\text{CFS},D} \cup N_D$$

N_O — **Origins** Shipper origin doors.

$N_{\text{CFS},O}$ — **Origin CFS**

Container Freight Stations near port of loading. LCL cargo is consolidated (stuffed) into shared FCL containers here.

N_{POL} — **Ports of Loading**

Container terminals. Stuffed containers gate in and load onto vessel.

$N_{\text{POD},A}$ — **POD Arrival**

Discharge port arrival nodes. Container unloads.

$N_{\text{POD},E}$ — **POD Exit**

Customs cleared; container available for inland pickup.

$N_{\text{CFS},D}$ — **Destination CFS**

CFS near POD where containers are broken down (deconsolidation) and individual shipments are released per HBL.

N_D — **Destinations**

Consignee delivery doors.

$$A = A_{\text{PU}} \cup A_{\text{OC}} \cup A_{\text{OCEAN}} \cup A_{\text{DW}} \cup A_{\text{DC}} \cup A_{\text{FD}}$$

A_{PU} — **Pickup** (i, j) , $i \in N_O$, $j \in N_{\text{CFS},O}$. Truck pickup from origin to origin CFS.

A_{OC} — **Origin CFS to POL**

(i, j) , $i \in N_{\text{CFS},O}$, $j \in N_{\text{POL}}$. Stuffed container trucked from CFS to terminal. Includes fixed CFS build-up (consolidation) time.

A_{OCEAN} — **NVOCC sailing**

(i, j) , $i \in N_{\text{POL}}$, $j \in N_{\text{POD},A}$. One arc per scheduled NVOCC sailing (lane, vessel, departure week).

A_{DW} — **Dwell**

(p_A, p_E) , one per POD. Unloading + customs clearance.

A_{DC} — **POD to destination CFS**

(p_E, c) , $p_E \in N_{\text{POD},E}$, $c \in N_{\text{CFS},D}$. Container trucked to destination CFS.

A_{FD} — **Final delivery**

(c, d) , $c \in N_{\text{CFS},D}$, $d \in N_D$. After deconsolidation, individual shipments delivered by truck (FTL or LTL).

Remark 1 (Origin CFS as the consolidation node). The origin CFS is functionally critical for LCL: it is the only physical location where multiple shippers' cargo merges into one container. Modeling it as an explicit node permits: (a) the fixed consolidation dwell time to enter feasibility checks, (b) future modeling of CFS dock door capacity and labor constraints, (c) alternative-CFS routing for forwarders with multiple CFS facilities serving the same port. The destination CFS plays the dual role for deconsolidation.

Remark 2 (LCL vs. FCL graph difference). The Ocean FCL model has no CFS nodes — FCL containers move directly from shipper door to port, and from port to consignee door. LCL inserts CFS nodes at both ends. This is a structural difference, not a parametric one.

2.2 Commodity-Specific Subgraph $G(N_k, A_k)$

The subgraph $G(N_k, A_k) \subseteq G(N, A)$ contains only arcs that are time-feasible for shipment k and lie on at least one complete $o(k) \rightarrow d(k)$ path. Construction algorithm parallels Ocean FCL (§??) with CFS dwell time added at both ends.

3 Sets and Indices

Symbol	Description
K	Set of LCL shipments (commodities), indexed by k
N, A	All nodes and arcs
$A_{PU}, A_{OC}, A_{OCEAN}, A_{DW}, A_{DC}, A_{FD}$	Arc subsets by type
$A^k \subseteq A$	Feasible arc set for commodity k
$A_S^k \triangleq A^k \cap A_S$	Feasible arcs of type S for commodity k
N_k	Nodes incident to A^k
S	Set of NVOCC sailings; each $s \in S$ has (POL, POD, ETD, ETA, CFS-cutoff)
$S^k \subseteq S$	Sailings feasible for commodity k (time-feasible from CFS cutoff to deadline)
T	Set of container types: $T = \{\text{FEU}, \text{TEU}\}$
C_s	Set of <i>potential</i> container slots on sailing s , indexed by c . Pre-allocated $ C_s \leq N_{\max}(s)$ where $N_{\max}$ is the sailing's container capacity.
$P(s)$	(POL, POD) pair served by sailing s
HAZ_k	Hazmat class of shipment k ($\emptyset$ if general cargo)
$\text{compat}(k, k')$	Boolean: shipments k and k' can share a container (hazmat segregation)

4 Parameters

4.1 Commodity Parameters

Parameter	Symbol	Description	Unit
Volume	v_k	Cargo volume (sum of pieces)	CBM
Weight	w_k	Gross weight	kg
Piece dimensions	D_k	Max single-piece L×W×H	cm
Cargo ready	t_k^{rdy}	Earliest pickup from origin	datetime
Deadline	T_k^{dead}	Latest acceptable arrival at destination	datetime
Origin	$o(k)$	Source node $\in N_O$	—
Destination	$d(k)$	Sink node $\in N_D$	—
Cargo type	τ_k	General / DGR / temperature / valuable	categorical
HS code	hs_k	6-digit Harmonized System code	—
Incoterm	inco_k	EXW / FOB / CIF / DDP	—
Budget cap	B_k	Hard cost ceiling; ∞ if not set	USD

W/M (weight/measurement) chargeable basis. While not used as an MVP optimization variable (the carrier-facing cost is per-container), the W/M chargeable basis is computed for

reporting and pricing:

$$WM_k \triangleq \max(v_k, w_k/1000) \quad [\text{CBM-equivalent}] \quad (1)$$

1 metric tonne (1000 kg) is treated as 1 CBM in maritime LCL convention. The shipment is “volume-rated” if $v_k \geq w_k/1000$ and “weight-rated” otherwise.

4.2 Container Type Specifications

Type	Code	Usable volume V_t	Payload W_t	Reference
40'HC	FEU	76 CBM	26,500 kg	MVP baseline (matches Ocean FCL)
20' std	TEU	33 CBM	24,000 kg	Selected when residual fits one TEU

Fill rate cap: $\eta = 0.97$. No real container packs to 100% — irregular cargo shapes and dunnage requirements consume capacity.

4.3 Pickup, CFS, Inland Ground Arc Parameters

For each ground arc $(i, j) \in A_{\text{PU}} \cup A_{\text{OC}} \cup A_{\text{DC}} \cup A_{\text{FD}}$:

Parameter	Symbol	Description	Unit
Cost	c_{ij}^G	Truck cost per shipment	USD
Mean transit	μ_{ij}^G	Mean transit time	days
Std dev	σ_{ij}^G	Std dev of transit	days
CFS consolidation time	β^O	Fixed origin CFS dwell (cargo arrival to container ready)	days
CFS deconsolidation time	β^D	Fixed destination CFS dwell (container arrival to cargo released)	days

MVP defaults: $\beta^O = 1.5$ days, $\beta^D = 1.5$ days (configurable per CFS).

4.4 NVOCC Sailing Parameters

For each sailing $s \in S$:

Parameter	Symbol	Description	Unit
POL, POD	$\text{pol}(s), \text{pod}(s)$	Origin and destination ports	—
NVOCC carrier	$\text{nv}(s)$	NVOCC operating this sailing	—
Underlying ocean carrier	$\text{oc}(s)$	Vessel operator (MSC, CMA CGM, etc.)	—
ETD	ETD_s	Scheduled departure	datetime
ETA	ETA_s	Scheduled arrival at POD	datetime
Mean transit	μ_s^{OC}	$\text{ETA}_s - \text{ETD}_s$	days
Std dev	σ_s^{OC}	Transit time variability	days
CFS cutoff	CFC_s	Latest time stuffed container must be at POL	datetime
Container capacity (FEU)	$N_{\max}^{\text{FEU}}(s)$	Max FEU containers sold on sailing s	containers
Container capacity (TEU)	$N_{\max}^{\text{TEU}}(s)$	Max TEU containers sold on sailing s	containers
FEU cost	κ_s^{FEU}	All-in cost to NVOCC for one FEU on sailing s	USD
TEU cost	κ_s^{TEU}	All-in cost to NVOCC for one TEU on sailing s	USD

Remark 3 (Container cost includes surcharges). κ_s^t is the NVOCC’s all-in carrier-facing cost: base ocean freight + BAF/EBS + LSS + ISPS + GRI on this sailing. Container yard fees and CFS fees are billed separately by node.

4.5 Hazmat Compatibility

For each pair of shipments (k, k') , compatibility is a Boolean parameter:

$$\text{compat}(k, k') = \begin{cases} 1 & \text{if } k, k' \text{ may share a container under IMDG segregation rules,} \\ 0 & \text{otherwise.} \end{cases} \quad (2)$$

In MVP, compat is a simple pairwise table derived from the IMDG class of each shipment. Full segregation rules (e.g., minimum distance between incompatible classes within a container) are deferred to P1.

5 Decision Variables

Variable	Type	Description	Domain
x_{ij}^k	Binary	1 if shipment k routed on arc $(i, j) \in A^k$	$\{0, 1\}$
$y_{k,c,s}$	Binary	1 if shipment k is loaded into container $c \in C_s$ on sailing s	$\{0, 1\}$
$z_{c,s,t}$	Binary	1 if container slot c on sailing s is used as type $t \in T$	$\{0, 1\}$
$t_k(i)$	Continuous	Arrival time of shipment k at node $i \in N_k$	$\mathbb{R}_{\geq 0}$

Container slot indexing. The set C_s pre-allocates an upper bound on containers per sailing. Practical bound: $|C_s| = N_{\max}^{\text{FEU}}(s) + N_{\max}^{\text{TEU}}(s)$, or $\lceil \sum_k v_k / (\eta V_{\text{TEU}}) \rceil$, whichever is smaller. Unused slots simply have $z_{c,s,t} = 0$ for all t .

6 Constraints

P.1 — Flow Conservation

For each commodity k and node $n \in N_k$:

$$\sum_{j:(n,j) \in A^k} x_{nj}^k - \sum_{i:(i,n) \in A^k} x_{in}^k = \begin{cases} +1 & n = o(k) \\ -1 & n = d(k) \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

P.2 — Shipment Assignment to Container

Each shipment is loaded into exactly one container slot across all feasible sailings:

$$\sum_{s \in S^k} \sum_{c \in C_s} y_{k,c,s} = 1 \quad \forall k \in K \quad (4)$$

P.3 — Container Volume Capacity

For each sailing s and container slot $c \in C_s$:

$$\sum_{k \in K} v_k \cdot y_{k,c,s} \leq \eta \cdot \sum_{t \in T} V_t \cdot z_{c,s,t} \quad (5)$$

If no container type is selected ($\sum_t z = 0$), the RHS is 0, forcing $y_{k,c,s} = 0$.

P.4 — Container Weight Capacity

$$\sum_{k \in K} w_k \cdot y_{k,c,s} \leq \sum_{t \in T} W_t \cdot z_{c,s,t} \quad (6)$$

P.5 — Single Type per Container Slot

$$\sum_{t \in T} z_{c,s,t} \leq 1 \quad \forall c \in C_s, s \in S \quad (7)$$

P.6 — Sailing Container Capacity

For each sailing s and container type t :

$$\sum_{c \in C_s} z_{c,s,t} \leq N_{\max}^t(s) \quad (8)$$

P.7 — Arc-to-Sailing Linkage

For each shipment k and ocean arc $(i, j) \in A_{\text{OCEAN}}^k$ corresponding to sailing s :

$$x_{ij}^k = \sum_{c \in C_s} y_{k,c,s} \quad (9)$$

If shipment k uses sailing s 's arc, it must be in exactly one container on that sailing.

P.8 — Hazmat Compatibility (Co-Loading Exclusion)

For each container slot $c \in C_s$ and each incompatible pair (k, k') with $\text{compat}(k, k') = 0$:

$$y_{k,c,s} + y_{k',c,s} \leq 1 \quad (10)$$

Incompatible shipments cannot share a container.

P.9 — CFS Cutoff

For each shipment k and each ocean arc $(i, j) \in A_{\text{OCEAN}}^k$ with sailing s :

$$t_k(i) \leq \text{CFC}_s + M \cdot (1 - x_{ij}^k) \quad (11)$$

The shipment's arrival at POL must precede the NVOCC's CFS cutoff.

P.10 — Arrival Time Propagation (Ground)

For each ground arc $(i, j) \in A_{\text{PU}}^k \cup A_{\text{OC}}^k \cup A_{\text{DC}}^k \cup A_{\text{FD}}^k$:

$$t_k(j) \geq t_k(i) + \tilde{\mu}_{ij}^G - M(1 - x_{ij}^k) \quad (12)$$

where $\tilde{\mu}_{ij}^G = \mu_{ij}^G + \beta^O$ for A_{OC} , $\mu_{ij}^G + \beta^D$ for A_{DC} , else μ_{ij}^G .

P.11 — Arrival Time Propagation (Ocean)

For each ocean arc $(i, j) \in A_{\text{OCEAN}}^k$ with sailing s :

$$t_k(j) \geq \text{ETA}_s - M(1 - x_{ij}^k) \quad (13)$$

P.12 — Deadline

$$t_k(d(k)) \leq T_k^{\text{dead}} \quad (14)$$

P.13 — Cargo Type Compatibility with Sailing

For each shipment k with $\tau_k = \text{DGR}$ and each ocean arc on sailing s :

$$x_{ij}^k \leq \text{dgr}(s) \quad (i, j) \in A_{\text{OCEAN}}^k, s = s(i, j) \quad (15)$$

P.14 — Piece Dimensions Fit Container

For each shipment k and container slot c where D_k does not fit in V_t for any $t \in T$:

$$y_{k,c,s} = 0 \quad (\text{pre-processing exclusion}) \quad (16)$$

Pieces longer than $\sim 12\text{m}$ or taller than $\sim 2.6\text{m}$ do not fit either FEU or TEU.

P.15 — Budget Cap (optional)

For shipments with $B_k < \infty$:

$$\text{cost}_k \leq B_k \quad (17)$$

where cost_k is shipment k 's allocated share of total cost (see Objective).

P.16 — Domain

$$x_{ij}^k, y_{k,c,s}, z_{c,s,t} \in \{0, 1\}, \quad t_k(i) \in \mathbb{R}_{\geq 0} \quad (18)$$

7 Objective Function

Minimize total cost across all shipments:

$$\begin{aligned} \min \quad & \sum_{k \in K} \sum_{(i,j) \in A_{PU}^k \cup A_{OC}^k \cup A_{DC}^k \cup A_{FD}^k} c_{ij}^G \cdot x_{ij}^k && \text{(ground cost)} \\ & + \sum_{s \in S} \sum_{c \in C_s} \sum_{t \in T} \kappa_s^t \cdot z_{c,s,t} && \text{(container cost (NVOCC pays per container, not per CBM))} \\ & + \sum_{k \in K} \text{surcharges}(k) && \text{(per-shipment surcharges (THC, ISPS, etc.))} \end{aligned}$$

Per-shipment cost allocation (for budget P.15 and reporting). The shipment's allocated share of container cost is proportional to its W/M chargeable basis within the container:

$$\text{cost}_k^{\text{ocean}} = \frac{\text{WM}_k}{\sum_{k': y_{k',c,s}=1} \text{WM}_{k'}} \cdot \kappa_s^t \cdot z_{c,s,t} \quad \text{where } y_{k,c,s} = 1 \quad (19)$$

This is a post-solve allocation, not a constraint. For the budget cap P.15, an MVP simplification is to bound the allocated cost_k by B_k ; full per-shipment cost linearization is deferred.

8 Sequential Decomposition Solution Strategy

The joint model above has $O(|K| \cdot |S| \cdot |C_s|)$ binary y variables and $O(|S| \cdot |C_s|)$ container-type z variables. At production scale ($|K| = 50\text{--}200$ LCL shipments per batch, $|S| = 10\text{--}30$ sailings per lane, $|C_s| = 10\text{--}30$ potential containers), the joint problem becomes prohibitive: $10^5\text{--}10^6$ binary variables and similar number of bin-packing constraints.

Two solving strategies are documented:

8.1 Strategy A — Joint MILP (correctness reference, small batches)

Solve the joint model as specified above. Tractable for $|K| \leq \sim 50$ shipments per batch with a 10-minute time limit. Use as the correctness reference and for small-batch production use.

8.2 Strategy B — Sequential Decomposition (production scale)

Decompose into two stages:

Stage 1 — Routing without bin-packing. Solve a relaxed problem treating sailing capacity as aggregate CBM and weight, not as discrete containers. Each sailing s has aggregate capacity:

$$V_s^{\text{agg}} = \eta \cdot V_{\text{FEU}} \cdot N_{\text{max}}^{\text{FEU}}(s) + \eta \cdot V_{\text{TEU}} \cdot N_{\text{max}}^{\text{TEU}}(s)$$

Aggregate weight cap defined similarly. The decision is shipment-to-sailing assignment only. This is a simple transportation LP with $O(|K| \cdot |S|)$ variables — solves in seconds.

Stage 2 — Bin-packing per sailing. For each sailing s that received shipments in Stage 1, solve a separate bin-packing MILP: choose container types and assign shipments to containers within the sailing. This is a variable-size bin-packing problem with at most ~ 30 shipments per sailing — solves in seconds with a standard MILP solver.

Solution coupling. Stage 1 may produce a solution that is infeasible in Stage 2 (cannot bin-pack the assigned shipments into the sailing’s container capacity). In this case:

1. Identify the over-loaded sailing.
2. Add a Benders-style feasibility cut to Stage 1: at least one shipment from the over-loaded sailing must be reassigned.
3. Re-solve Stage 1.

Typical convergence: 2–5 iterations. Total solve time at production scale: ~ 1 –3 minutes.

Optimality gap. Sequential decomposition is an approximation. Empirically (literature on bin-packing-aware LCL consolidation), the optimality gap relative to the joint solution is 2–8% on realistic instances, with smaller gaps on tighter-capacity sailings.

When to use which.

- $|K| \leq 50$ per batch: solve joint MILP directly.
- $|K| > 50$: use sequential decomposition. Joint solver used only for spot-check validation on small representative subsets.

9 Linearization and Tightening

9.1 Big-M Tightening on Time Constraints (P.9, P.10, P.11)

Use the tightest valid bound: $M = T_k^{\text{dead}} - t_k^{\text{rdy}} + \epsilon$.

9.2 Symmetry Breaking on Container Slots

Container slots within a sailing are interchangeable, creating symmetry that slows LP relaxation. Break by ordering:

$$z_{c,s,t} \leq z_{c-1,s,t} \quad \forall c \geq 2, s, t \quad (20)$$

Forces filling slot 1 before slot 2 of the same type on the same sailing.

9.3 Valid Inequality — Minimum Container Count

For each sailing s :

$$\sum_{c,t} z_{c,s,t} \geq \left\lceil \frac{1}{\eta V_{\text{FEU}}} \sum_k v_k \cdot \left(\sum_{c \in C_s} y_{k,c,s} \right) \right\rceil \quad (21)$$

At least enough containers to hold the total assigned volume. Tightens LP relaxation. This is a valid inequality (cut), not a constraint required for correctness, but materially improves solving time.

10 LCL vs. FCL — Model Comparison

Aspect	Ocean FCL (<code>ocean_fcl_routing.tex</code>)	Ocean LCL (this document)
Graph nodes	4 layers (O, POL, POD, D)	6 layers (O, CFS-O, POL, POD, CFS-D, D)
CFS dwell	—	Explicit at both ends
Capacity	String allocation (TEU slots)	Container slots per sailing
Container mix	Per shipment (pre-computed)	Per consolidated container (decision)
Consolidation	—	Joint bin-packing $\times$ routing
Hazmat	Per shipment	Pairwise compatibility on co-loading
Variables	$O(K \cdot A)$	$O(K \cdot S \cdot C_s)$
Tractability	200+ shipments comfortable	50 joint; 200+ via decomposition

11 Open Items and Future Extensions

1. **Probabilistic transit time objective.** MVP uses deterministic mean transit and a hard deadline. P1: $\Pr(\text{arrival} \leq T_k^{\text{dead}}) \geq p$.
2. **Full 3D bin-packing.** MVP uses aggregate volume cap with $\eta = 0.97$. P1: 3D bin-packing for awkward piece dimensions and stackability.
3. **Multi-leg NVOCC routing.** MVP single-leg only. P1: NVOCC transshipment through hub CFS (e.g., Singapore consolidation hub for SE Asia $\rightarrow$ Europe).
4. **Full IMDG hazmat segregation.** MVP pairwise compatibility table. P1: full IMDG segregation distances within container, ventilation, dunnage.
5. **CFS dock door capacity.** MVP assumes infinite throughput. P1: model dock door and labor capacity at CFS.
6. **NVOCC vs. self-consolidation.** MVP treats us as the NVOCC. Some forwarders buy LCL capacity from third-party NVOCCs at a per-CBM rate. P1: support both modes as separate arc types.
7. **Consolidation across LCL and FCL.** MVP solves LCL separately from FCL. P1: joint LCL/FCL optimizer where a partial-load FCL shipment can be downgraded to LCL consolidation if cheaper.
8. **Multi-shipper LCL aggregation across batches.** MVP optimizes each batch independently. P1: rolling-horizon consolidation where a shipment may be held to consolidate with a later batch if total cost decreases.
9. **Per-shipment cost as linearized constraint (for P.15 budget).** Current per-shipment cost allocation is post-solve. P1: linearize for hard budget enforcement.
10. **Equipment availability at CFS.** MVP assumes empty FEU/TEU containers always available at the origin CFS. P1: model equipment stock and repositioning.